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A-MASA: Arabic Multi-Domain Aspect-Based Sentiment Analysis Datasets

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Abstract

The rapid growth of Natural Language Processing (NLP) applications has gained interest in various languages, including Arabic. One critical NLP task is Aspect-Based Sentiment Analysis (ABSA). ABSA involves identifying the sentiment expressed toward specific aspects or attributes of entities mentioned in a piece of text, rather than determining the overall sentiment. Despite the increased use of Semitic languages like Arabic in NLP over the past decade, there remains a lack of resources and datasets in Arabic, particularly in dialectical Arabic. Additionally, existing ABSA datasets are often limited to a single domain, which does not support all types of ABSA, such as multi-domain ABSA, nor do they allow for testing a model's generalization capabilities. This study addresses these limitations by constructing multi-domain ABSA datasets in Arabic for the tasks of aspect extraction and aspect polarity detection. The datasets include both real and synthetic data, covering dialectical and modern standard Arabic, and comprising 6,500 records across five domains. These datasets can be utilized in single-domain, cross-domain, and multi-domain experiments for ABSA. Furthermore, we evaluated the constructed datasets using various transformer-based models and assessed the impact of generating synthetic data on the diversity and balance of the data, as well as on model performance. The results reported in this research demonstrate that the data is valid for use in ABSA tasks. Moreover, the models used for evaluation achieved competitive or superior results compared to existing models on various datasets. Additionally, the annotation guidelines we created are presented as recommended practices applicable to different tasks. All datasets used in this study are publicly available for research purposes.

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Keywords: Aspects Extraction; Arabic Aspect-Based Sentiment Analysis; Natural language processing; Datasets.

1. Introduction

Researchers have explored several levels of Sentiment Analysis (SA), including document-level, sentence-level (short document), and aspect-level, also known as Aspect-Based Sentiment Analysis (ABSA) [1]. Document-level

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and sentence-level SA face challenges when handling phrases with multiple aspect terms and varying sentiments. Conversely, ABSA offers a more detailed and efficient analysis of each aspect in a given sentence [2]. Traditional SA assigns a single score for the whole sentence [3], whereas ABSA identifies sentiment regarding each aspect in a given sentence. According to [17], there are four tasks shared in the SemEval workshop. These tasks are task 1 (T1): Aspect/opinion term entity (OTE) extraction. Task 2 (T2): OTE polarity classification. Task 3 (T3): Aspect category extraction. Task 4 (T4): Aspect category polarity classification. ABSA has gained significant attention from researchers in recent years. However, the Arabic language has not fully leveraged this progress due to the limited availability of datasets. Additionally, ABSA has gained traction in the SemEval shared task since 2014 [4]. Aspect extraction can be defined as a multi-class classification task where each word in a sentence is classified as an aspect term or not. Different labeling schemes can be used for this text classification task.

Arabic is the official language in 22 nations and is utilized by over 400 million individuals globally [5]. There are many challenges for the Arabic language in performing ABSA tasks. According to Internet World Stats [7], Arabic ranks fourth in terms of internet usage, with over 230 million users. Over the past decade, SA in Arabic text has gained considerable attention from researchers. Nonetheless, the volume of published studies remains relatively low compared to those in English, particularly concerning aspects such as ABSA. The most common challenge is the availability of Arabic annotated datasets, which significantly impacts the richness of research in the Arabic region. Recently, the Arabic language has witnessed significant growth due to the essential use of the Internet daily. There are many resources released for SA tasks in the Arabic language. Moreover, many researchers have contributed to ABSA; however, the datasets utilized in these studies primarily emphasize popular domains such as hotels, books, and news [6], [8], [13], [15]. Unfortunately, most of the remaining datasets are not publicly accessible.

The main contribution of this research is the creation of multi-domain datasets for ABSA in the Arabic language. In this study, reviews from five domains were collected from different resources. Real data and synthetic data in all these domains were presented in both dialectical and modern standard Arabic. The domains covered in these datasets are tourism, products, movies, restaurants, and telecommunication. The first four domains were collected from a previously published dataset and annotated manually for the task of ABSA instead of the SA task. Additionally, data were collected from various social media platforms besides an AI-based chatbot was used to generate synthetic reviews. The constructed datasets were evaluated using different transformer-based models. The highest accuracy was achieved in the movies domain with a score of 97.73%, while the products domain attained an F1 score of 65.53%. Furthermore, the effect of generating a synthetic dataset on both the diversity and balance of the data, as well as on the model's effectiveness, was evaluated.

The paper is organized as follows: An overview of the literature review, including recent datasets in the field, is presented in Section 2. The dataset construction pipeline is illustrated in Section 3. Section 4 discusses the dataset setup and the conducted experiments in detail. Section 5 discusses the results of tested architectures. Finally, Section 6 concludes the paper and suggests future work.

2. Literature Review

Recently, many researchers have focused on the problem of limited data resources in Arabic, particularly for SA. Additionally, several datasets are available for ABSA in Arabic, however, most of these resources have at least one of the following limitations: they cover only one domain, are not publicly available, or have limited size [6]. One of these datasets is called HAAD, a Modern Standard Arabic (MSA) dataset comprising 1,513 book reviews in Arabic [13]. This dataset covers aspect terms and categories with polarities. The authors evaluated the dataset using baseline models and achieved an F1 score of 0.233. An ontology-based model reported in [14] achieved an F1 score of 42.7%. Another domain covered in [15] is news posts written in MSA too. In this dataset, posts were automatically crawled from different news networks covering T1 and T2. About 20% of the collected data was selected for manual annotation using the BRAT rapid annotation tool[1]. The resulting dataset consists of 2,265 news posts in XML format for all tasks related to ABSA. The baseline model for the aspect extraction task achieved an F1 score of 39.4%. Another lexicon-based model evaluated by the same authors achieved an F1 score of 51.4%. In [16], the same dataset was trained on

[1] http://brat.nlplab.org/

multiple transformer-based models, with the highest results achieved by the AraBERT+BiGRU+CRF model, which attained an F1 score of 88.1%.

In [8], the authors collected around 15,562 hotel reviews from well-known hotel booking websites written in both MSA and multiple Dialectical Arabic (DAs). A subset of 2,291 reviews was selected for manual annotation for T1 and T2 according to the guidelines provided in [8]. The Support Vector Machine (SVM) model used for evaluating this dataset achieved an F1 score of 30.9%. Another work leveraged a deep LSTM encoder followed by a logistic regression layer to train this dataset using Word2vec embeddings, achieving an F1 score of 47.3% [10]. In [9], the dataset was evaluated in a multilingual training scenario and achieved an F1 score of 52%. A Character-level BiLSTM with a CRF layer was introduced in [11] and has achieved an F1 score of 69.98%. Moreover, the (GRU) model demonstrated high performance in [12], achieving an F1 score of 93%. The ALR dataset [18] is an Arabic laptop review collection from different websites. This dataset contains 1,028 reviews written in both MSA and DA for T3 and T4. The SVM model used for training and testing these reviews, along with n-gram features embeddings, achieved an F1 score of 0.315 for the category detection task. For the category sentiment analysis, the model achieved an accuracy of 0.732. In [22], the authors constructed a unique MSA and DA dataset by annotating 2,071 mobile application government reviews in Arabic. With the help of a special application named "GARSA," the authors accomplished the annotation process [23].

The authors of [19] built a dataset for airline tweets using the Twitter API tool [2]. They focused on solving T3 and T4 tasks using the SVM classifier, along with AraVec and FastText word embedding techniques. In a later work, the authors of [21] presented a dataset of food delivery reviews consisting of 4,879 tweets collected using keywords related to food delivery services. The GigaBERT and AraBERT models each achieved an F1 score of 0.77 for T1. As shown, the number of available ABSA datasets is insufficient compared to other tasks, and the covered domains remain limited. In this study, we manually annotated and evaluated multi-domain ABSA datasets for T1 and T2 in the ABSA task in Arabic. These datasets were evaluated using various transformer-based models, achieving competitive results compared to other models.

3. Datasets Construction

Existing datasets typically focus on single domains, limiting their utility in assessing model generalization. According to [25], manual data annotation techniques are superior in efficiency and accuracy compared to other approaches. However, the main disadvantage of this method is its substantial time and resource consumption, which restricts the ability to manually annotate more reviews in our case. Hence, this research focuses on building multi-domain datasets in both MSA and DA for T1 and T2 in ABSA. The methodology for creating these datasets includes four phases: data collection, data cleaning, data annotation, and data preparation for training. A pipeline for the entire process is provided in Fig.1. The domains covered in this work are tourism (ATT), products (PROD), restaurants (RES), movies (MOV), and telecommunications (TELC). The annotated reviews were formatted as comma-separated value (CSV) files for easier conversion to Python [3] data frames later.

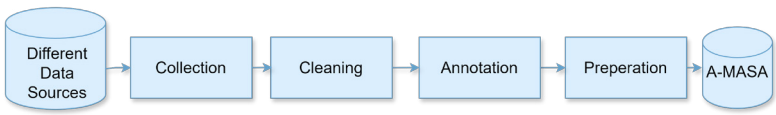

Fig. 1. Dataset Construction Pipeline.

[2] https://developer.x.com/en/docs/twitter-api
[3] https://www.python.org/

3.1. Dataset Collection

This research's data sources include real and synthetic data across five domains, as described in Fig. 2. The data from the first four domains, which are all real, come from previously published datasets comprising a total of 33,000 annotated reviews [24]. These reviews cover content related to movies, hotels, restaurants, and products. They were automatically scraped from various websites using the open-source Scrapy framework[4] for building custom web crawlers. We manually re-annotated 1,500 reviews for ABSA instead of SA from each domain, except for the MOV domain, where 500 reviews were selected due to their longer sentences. Details of each domain's data are provided in Table 1. In the TELC domain, over 2,000 tweets about Egyptian telecom companies were collected using the Twitter API [28] and filtered, resulting in 32% of the data. Additionally, approximately 30,000 comments were scraped from Facebook using the Selenium tool[5], and after data filtration, these comments also accounted for 32%. Most of these reviews were irrelevant to telecommunications, as Arabic telecommunication reviews on social media frequently express anger without specifying complaints in detail. To improve data quality in this domain, more data is needed. One effective technique for data generation is using generative AI chatbots. In this study, we utilized Gemini chatbot [26] to generate synthetic data, covering 36% of the TELC domain. The prompt used for synthetic data generation can be found in Appendix Appendix A.

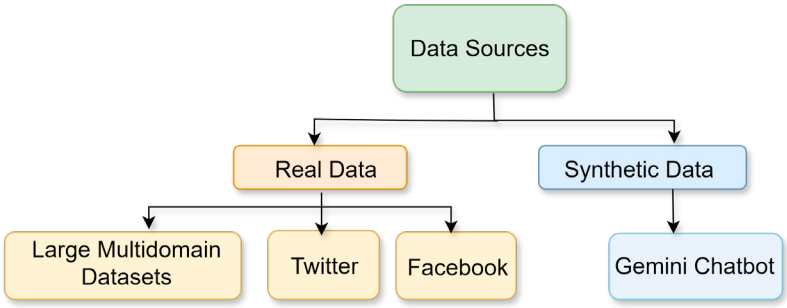

Fig. 2. Data Sources.

3.2. Dataset Cleaning

To ensure data quality, the collected data underwent rigorous cleaning following established procedures documented in the literature [19], [20], [24], [33] and [18]. These steps include removing URLs, and redundant, irrelevant, meaningless, and empty records. Additionally, we experimented with various text-stemming techniques, leading us to develop our methodology. This approach ensures text normalization between the original reviews and the extracted aspects, facilitating accurate data labeling.

3.3. Dataset Annotation

In English language, many annotation guidelines are provided by the SemEval 2015 Task [17], and most studies adhere to these standards. However, no specific guidelines exist for Arabic. To address this gap, we conducted an extensive analysis of existing Arabic reviews, aligning with SemEval guidelines to develop our annotation standards. For our extracted annotation guidelines, see Appendix D. Ten annotators, native Arabic speakers with relevant domain expertise, participated in the annotation process. The average age of annotators was 20 years old, with approximately 60% male and 40% female participants, all from Egypt. The annotation process progressed through several phases,

[4] https://scrapy.org/
[5] https://www.selenium.dev/

with annotations evaluated for quality assurance by determining the majority consensus among three annotators per domain. This study includes annotated datasets for T1 and T2. T1 labels are B (beginning of aspect term), I (inside aspect term), or O (not an aspect term). T2 categorized each extracted aspect term into positive, negative, or neutral sentiment classes. The annotation process underwent revisions using both manual and automated approaches. An automated method involved comparing manually extracted aspects to the original aspects in each sentence. Additionally, the number of extracted aspects was compared to the number of aspect sentiment classes to minimize potential errors.

3.4. Dataset Preparation

To prepare the datasets for training, the data is formatted according to the IOB2 scheme [29]. This scheme labels the first word in an aspect term as B, subsequent words in the term as I, and words not part of an aspect term as O. Aspect extraction follows a token classification task, aiming to classify each word in the input sentence into one of these classes: I, O, or B. An example of IOB2 labeling for the ATT domain is illustrated in Appendix Appendix E. This process was automated using a Python script.

3.5. Datasets Statistics

An overview of the domains is provided in table 1. As shown in the table, the data is unbalanced, necessitating specific preprocessing when feeding it into models. The RES domain exhibits the highest number of aspects, reflecting extensive opinions about food. The tourism domain follows with the second highest number of aspects. The distribution of sentiment classes across each domain is visualized in Appendix C. Additionally, the number of tokens or single words in each domain is represented in the same appendix. All datasets are available on GitHub for research purposes [6].

Table 1. Datasets statistics.

Domain	# Reviews	# Tokens	# Aspects	# Neutral Aspects	# Positive Aspects	# Negative Aspects
ATT	1500	55694	4406	154	4288	300
PROD	1500	17932	1430	29	1172	258
MOV	500	49270	974	16	725	249
RES2	1500	51835	6116	496	5225	891
TELC	1500	18849	815	91	627	188

4. Experiments

4.1. Dataset Setups

After cleaning and labeling the data in IOB2 format, it is tokenized using appropriate tokenizers tailored for each model. Specifically, the AraBERT model [27], MARBERT [32], and QARiB [7] were employed, each with its corresponding tokenizer. Two special symbols, [CLS] and [SEP], were added to the tokens: [CLS] marks the beginning of each input sentence, while [SEP] serves as a separate [27]. The [CLS] token, which encapsulates information about the input sentence, was fed into the model for predicting class labels in the model.

4.2. Models

Models utilized for ABSA encompass rule-based, Machine Learning (ML), Deep Learning (DL) [31], transformer-based, and LLMs (Large Language Models). Transformer-based models have recently shown significant efficacy

[6] https://github.com/YomnaEid/A-MASA-Arabic-Multidomain-Aspect-based-Sentiment-Analysis-Datasets.git
[7] https://github.com/qcri/QARIB

across various Arabic language tasks [30]. The evaluation in this study employs AraBERT, MARBERT, and QARiB. AraBERT was chosen as one of the baseline models due to its established performance in Arabic text processing. MARBERT was pre-trained on Twitter data containing text in both MSA and DA. QARIB was pre-trained similarly to MARBERT on both MSA and various DA. The MSA text sources include news and movie/TV subtitles, while the DA text comprises tweets. Each of these transformers was followed by a linear layer and subsequently, a softmax layer to predict labels (B, I, O) for each token. Each model was fine-tuned on each dataset domain separately, using 80% for training and 20% for testing, aiming for improved domain-specific predictions. Python and PyTorch [8] tools were utilized for implementation. The AdamW optimizer was selected to facilitate convergence and enhance performance during transformer model fine-tuning. We chose a learning rate of 2e-5, following common practices for fine-tuning pre-trained transformers. Experimentation with different numbers of epochs revealed that the optimal range was between 15 and 20 epochs, varying by domain. Using more epochs often resulted in overfitting due to the dataset's relatively small size. After testing various options, we settled on a batch size of 64.

5. Results and Discussion

The datasets were evaluated using three transformer-based models: AraBERT, MARBERT, and QARiB. Detailed results are presented in Table 2. In the experiments, the MOV domain achieved the highest accuracy of 97.73% using the QARiB transformer-based model. Similarly, the highest F1-score 69.50% is achieved in the TELC domain, which was also by the QARiB model. The QARiB-based model achieved the highest F1 score among the fifteen experiments conducted. This model was selected for detailed analysis using synthetic data in the TELC domain. To evaluate the impact of synthetic data generation, a separate experiment is presented in Table 3. This table presents a comparison among three types of datasets: the first using real data exclusively, the second using synthetic data alone, and the third combining both real and synthetic data. The synthetic dataset alone did not significantly enhance the model's performance. However, combining both real and synthetic data resulted in a notable improvement, increasing accuracy by more than 8% and F1 score by 6% compared to using real data alone.

This superior performance can be attributed to the enhanced diversity in the data. Synthetic data generation balanced and enriched the dataset by increasing the dataset size. The richness and quality of aspect terms in each domain significantly affect model performance; aspect extraction models tend to perform better in domains with a larger number of well-represented aspects. Furthermore, combining real and synthetic data exposed the model to a broader spectrum of examples, thereby enhancing its ability to generalize to new, unseen data. The process of synthetic data generation helped overcome the limitations of the initial data collection including unrelated and spam data. Moreover, table 4 compares the performance of existing models used for T1 across different domains with our models and annotated domains. As shown, the achieved F1 score is comparable to, and in some cases higher than, those reported in other studies. The evaluation metrics used in this research are Accuracy and F1 score, with the F1 score being macro-averaged. For detailed equations of these metrics, see Appendix B.

We ensured dataset representativeness by encompassing five diverse domains, including both real and synthetic data, and covering content in both MSA and DA. It is noticed that the aspects classes are imbalanced, as well as the sentiment classes. The number of occurrences for class 'O' is much larger than for class 'B' or 'I'. Additionally, The number of positive reviews greatly surpasses that of negative reviews. This observation was similarly noted in [24] and [33]. This fact may require using additional techniques, for example using class weight. This research would benefit from more annotated data, but manual annotation is time-consuming and costly, and access to sufficient domain-specific data is limited. These constraints hindered dataset expansion, though we are actively exploring strategies to overcome these limitations and improve our data collection processes for future work. Moreover, the annotation guidelines we developed are presented as suggested approaches suitable for various tasks.

[8] https://pytorch.org/

Table 2. Datasets evaluation using three models.

Model	Domain	Accuracy	F1-Score
AraBERT	ATT	92.21%	43.15%
	RES	88.16%	32.87%
	PROD	89.74%	45.70%
	MOV	**97.72%**	32.95%
	TELC	88.98%	**50.04%**
MARBERT	ATT	92.17%	38.17%
	RES	89.20%	43.10%
	PROD	92.20%	**55.97%**
	MOV	**97.72%**	32.95%
	TELC	88.88%	39.56%
QARiB	ATT	92.51%	45.84%
	RES	89.31%	41.24%
	PROD	92.92%	65.53%
	MOV	**97.73%**	37.22%
	TELC	96.16%	**69.50%**

Table 3. Comparison between real, synthetic, and combined datasets in TELC domain using QARiB.

Data Type	Accuracy	F1-measure
Real	89.39%	56.64%
Synthetic	88.93%	63.85%
Combined	**96.16%**	**69.50%**

Table 4. Performance of different models across various domains.

Model	Other layers	Domain	F1
AraBERT [34]	CRF	HAAD	47.63%
AraBERT [A-MASA]	linear	TELC	50.04%
AraBERT [35]	LCF	Education	54%
AraBERT [16]	linear	News	**87.10%**
MARBERT [35]	LCF	Education	44%
MARBERT [A-MASA]	linear	PROD	**55.97%**
QARiB [35]	LCF	Education	44%
QARiB [A-MASA]	linear	TELC	**69.50%**

6. Conclusion and Future work

In this study, we introduced multi-domain ABSA datasets, covring T1 and T2, collected from diverse sources. While the majority of these datasets are real, 36% of the samples are synthetic. The datasets comprise reviews from domains such as tourism, movies, restaurants, products, and telecommunications, presented in both MSA and DA. We performed benchmarking experiments on each domain dataset using different transformer-based models. The QARiB model achieved the highest accuracy in the MOV domain and the highest F1-score in the TELC domain. These datasets can be used for single-domain, cross-domain, and multi-domain experiments. We also evaluated the impact of generating synthetic datasets on diversity, balance, and model performance. The conducted experiments showed that significantly enhanced the results significantly. The provides datasets in this study can serve various NLP tasks such as sentiment analysis, NER (Named Entity Recognition), dialect identification, and emotion causality. For future work, we plan to explore other types of text on social media platforms, such as news, blogs, and advertisements, believing that the nature of these texts will reveal hidden patterns that will significantly impact future experiments. Furthermore, additional tasks such as T3 and T4 can be targeted. Additionally, we plan to expand the dataset size by exploring more domains.

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Appendix A. Gemini Prompt

This appendix contains the details of the Gemini prompt sent to the Gemini chatbot.

Make a 50 review about telecommunication companies in Egypt like Etisalat using the following criteria: 1- make them in Arabic 2- no duplicates for the sentences (reviews) 3- don't make them long 4- it may contain aspects or not 5- make the reviews comprehensive 6- DONT MAKE ENGLISH TRANSLATION REMOVE THEM FORM THE SENTENCE 7- don't tell us the company name just make the sentence contain the review in Arabic 8- export all the reviews in a table to be used as CSV.

Fig. A.3. Gemini Prompot.

Appendix B. Evaluation Metrics Equations

This appendix contains the equations used for evaluation metrics.

B.1. Equation for accuracy:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad \text{(B.1)}$$

$$F1 - Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad \text{(B.2)}$$

Where TP is True Positive, TN is True Negative, FP is False Positive, FN is False Negative.

Appendix C. Domains Statistics

This appendix illustrates the introduced five domains statistics.

C.1. Sentiment Classes in Each Domain.

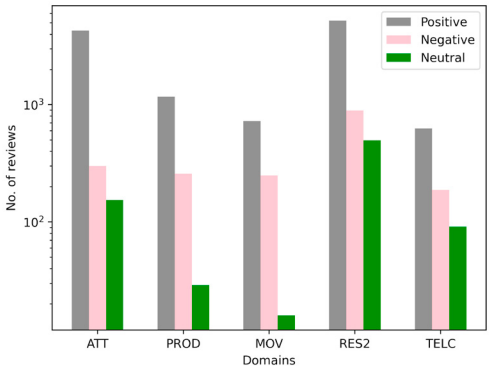

Fig. C.4. Sentiment Classes in Each Domain.

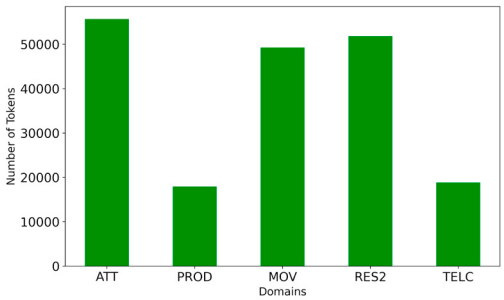

Fig. C.5. Number of Tokens Across Domains.

Appendix D. Annotation Rules.

This appendix contains our extracted annotation guidelines on the given domains.

Rules	Example Input	Example Output
1- If the whole review is written in English, remove the row.	Hi, good	remove
2- If the aspect term consists of multiple words, extract them all.	باب الفصل كبير	باب الفصل
3- If the sentence has multiple aspects, extract them all and separate them by '، '.	الاكل حلو بس الجو حر	الاكل، الجو
4- Avoid typos when extracting aspects, for example, 'شنطه' should be extracted as 'شنطة'.	شنطه	شنطه
5- If the aspect term has a suffix or prefix, like: 'ضمائر أو حروف عطف أو جر' => remove these suffix or prefix.	للفلسفة تهديداتها	فلسفة تهديدات
6- If the review consists of more than 5 lines, split it into multiple rows to get 5 lines maximum/ row.		
7- If words are concatenated without spaces, add them	المنتجحلو	المنتج حلو
8- For the 'TELC' domain only, if the review is not about telecommunication, remove it.	قسم اتصالات في هندسة صعب	remove
9- If the row is duplicated, remove it.		
10- The adjective may contain multiple words.	حديقة مليئة بالتماثيل التاريخية	حديقة، التماثيل
11- The adjective may come after the aspect and it should be extracted too.	أحسن خدمة	خدمة
12- There might be no aspects in the sentence.	ممتاز جدا	
13- There might be many words separating the aspect term and adjective.	كوبري استانلي هو من أجمل الأماكن	كوبري استانلي
14- The sentiment adjective might not be an adjective but rather a separable phrasal verb.	قضاء يوم في الحدائق من امتع التجارب	قضاء يوم في الحدائق

Fig. D.6. Annotation Guidelines

Appendix E. An Example of IOB2 scheme.

This appendix contains an example of IOB2 labeling scheme for an input sentence in Arabic.

Sentence	Aspects	IOB2 Labels
مكان جداً جميل وراقي السياحه البحريه ممتازه اسعارهم مناسبه ، اهم شي تركزون هنا علئ لحم الخنزير والكحول	مكان، السياحه البحريه، اسعارهم	B, O, O, O, B, I, O, B, O, O, O, O, O, O, O, O, O, O

Fig. E.7. An Example of IOB2 scheme.